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Fluid supply apparatus

Abstract

A fluid supply apparatus that can eliminate the need for a power supply source, and store fluid efficiently. The apparatus may comprise a first balloon that is formed of an elastic body to be capable of storing the fluid, a second balloon that is formed of an elastic body to be capable of storing the fluid, and has a larger contraction force than a contraction force of the first balloon, a support member that supports both of the first balloon and the second balloon, and a casing that houses the first balloon and the second balloon, and defines inflation ranges of the first balloon and the second balloon, in which the support member has a flow path formed to permit fluid communication between an interior of the first balloon and an interior of the second balloon.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2518145	12/1949	Jenson	222/386.5	B67D 3/00
5137175	12/1991	Kowalski	222/105	B67D 1/0462
9828170	12/2016	Nomura	N/A	B65B 31/003
10160592	12/2017	Sugawara	N/A	B65D 83/38
2006/0054634	12/2005	Mekata	222/105	B05B 11/1081
2009/0108021	12/2008	Hansen	222/105	C12N 15/8247

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2002017191	12/2001	JP	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit and priority of Japanese Application No. 2022-126827, filed Aug. 9, 2022. The entire disclosure of the above application is incorporated herein.

TECHNICAL FIELD

(2) The present invention relates to a fluid supply apparatus, and more particularly, to a fluid supply apparatus for supplying a liquid for rearing organisms such as mice.

BACKGROUND ART

(3) Recent years have seen experiments conducted for the purpose of studying the effects of a zero

gravity environment or a microgravity environment on organisms in outer space so as to rear organisms such as mice flown on a spacecraft such as a space shuttle in outer space. The rearing environment for mice or the like in outer space requires facilities for supplying mice or the like with water or liquid diet (hereinafter referred to as water or the like) in the same manner as in the case of rearing mice or the like on the ground (for example, see PTL 1). Examples of common facilities for supplying water or the like to mice or the like on the ground include an automatic fluid supply apparatus using gravity. However, such an automatic fluid supply apparatus using gravity cannot be used in a zero gravity environment.

(4) Thus, there has been proposed a device for automatically supplying water or the like using a motor-driven syringe pump or the like as facilities for supplying water or the like to mice or the like in outer space.

CITATION LIST

Patent Literature

(5) PTL 1: Japanese Patent Laid-Open No. 2002-017191

SUMMARY OF INVENTION

Technical Problem

(6) However, such a motor-driven device requires a power supply source and has a relatively heavy weight and large footprint. Therefore, it is not preferable to install such a motor-driven device in a spacecraft with limited loading weight and loading space. In addition, motor-driven devices may fail. A failure of the motor-driven device requires a repair by an astronaut and involves a time-consuming maintenance and management effort.

(7) An object of the present invention is to provide a fluid supply apparatus that can solve problems of the prior art described above, eliminate the need for a power supply source, and store fluid efficiently.

Solution to Problem

(8) The present invention provides a fluid supply apparatus for supplying stored fluid to outside comprising a first balloon that is formed of an elastic body to be capable of storing the fluid, a second balloon that is formed of an elastic body to be capable of storing the fluid, and has a larger contraction force than a contraction force of the first balloon, a support member that supports both of the first balloon and the second balloon, and a casing that houses the first balloon and the second balloon, and defines inflation ranges of the first balloon and the second balloon, wherein the support member has a flow path formed to permit fluid communication between an interior of the first balloon and an interior of the second balloon.

(9) In this case, the support member includes a connection portion that connects an end of the first balloon and an end of the second balloon, the ends being adjacent to each other, and the flow path may be formed in the connection portion. The support member is a tubular member, and may extend through the first balloon and the second balloon. The support member may extend from one end to the other end of the casing. The support member has a plate shape, and the first balloon may be attached to one surface of the support member and the second balloon may be attached to the other surface of the support member. The casing may be provided with a sensor for detecting that at least one of the first balloon and the second balloon has contacted the casing.

(10) The present invention provides a fluid supply apparatus for supplying stored fluid to outside comprising a first unit that comprises a first balloon that is formed of an elastic body to be capable of storing the fluid, a first support member that supports the first balloon, and a first casing that houses the first balloon and defines an inflation range of the first balloon, and a second unit that comprises a second balloon that is formed of an elastic body to be capable of storing the fluid and has a larger contraction force than a contraction force of the first balloon, a second support member that supports the second balloon, and a second casing that houses the second balloon and defines an inflation range of the second balloon, wherein the first support member and the second support member have a flow path formed to permit fluid communication between an interior of the first

balloon and an interior of the second balloon.
(11) In this case, the first casing may be provided with a sensor for detecting that the first balloon has contacted the first casing. The second casing may be provided with a sensor for detecting that the second balloon has contacted the second casing.
(12) In the above-described invention, the first balloon and the second balloon may be different from each other in thickness. The first balloon and the second balloon may be different from each other in inner diameter. The first balloon and the second balloon may be different from each other in length.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** illustrates a front view of a fluid supply apparatus according to a first embodiment of the present invention.
(2) FIG. **2** schematically illustrates a cross section of a shaft and balloons.
(3) FIG. **3**A schematically illustrates the fluid supply apparatus and sensors, and FIG. **3**B is a graph showing the change in an internal pressure with the change in elapsed time of discharge.
(4) FIG. **4** illustrates a perspective view of a fluid supply apparatus according to a second embodiment of the present invention.
(5) FIG. **5** illustrates a cross section of the fluid supply apparatus according to the second embodiment of the present invention.
(6) FIG. **6** illustrates a schematic perspective view through the fluid supply apparatus in which balloons are inflated.
(7) FIG. **7** illustrates a schematic sectional view of the fluid supply apparatus in which balloons are inflated.
(8) FIG. **8** illustrates a front view of a fluid supply apparatus according to a third embodiment of the present invention.
(9) FIG. **9** illustrates a sectional view of a balloon unit.
DESCRIPTION OF EMBODIMENTS
(10) Hereinafter, preferred embodiments of the present embodiment will be described with reference to the drawings.
First Embodiment
(11) FIG. **1** illustrates a front view of a fluid supply apparatus according to a first embodiment of the present invention. FIG. **2** schematically illustrates a cross section of a shaft and balloons, FIG. **3**A schematically illustrates the fluid supply apparatus and sensors, and FIG. **3**B is a graph showing the change in an internal pressure with the change in elapsed time of discharge.
(12) A fluid supply apparatus **100** according to the present embodiment is used, for example, in outer space to supply water or liquid diet (organism rearing liquid, hereinafter referred to as water or the like) to organisms such as mice to be reared in outer space. In particular, the fluid supply apparatus **100** according to the present embodiment may be mounted on a spacecraft for launch and/or recovery.
(13) As illustrated in FIG. **1**, the fluid supply apparatus **100** comprises a shaft **10** (support member), a first balloon **11**, a second balloon **12**, a first cap **13**, a second cap **14**, and a case (casing) **15**.
(14) The case **15** has a cylindrical shape formed of a transparent resin material. In the case **15**, the first cap **13** is incorporated at one end, and the second cap **14** is incorporated at the other end. The first balloon **11** and the second balloon **12** are supported by the shaft **10** supported by the first cap **13** and second cap **14**. An inner peripheral surface of the case **15** is formed so as to contact the first balloon **11** and the second balloon **12** inflated by injecting water or the like. Thus, the case **15** is adapted to define inflation ranges of the first balloon **11** and the second balloon **12**.

(15) The first balloon **11** is formed to be inflated into an apple-like shape in which a portion on the first cap **13** side is recessed in a longitudinal direction x, and the second balloon **12** is formed to be inflated into an apple-like shape in which a portion on the second cap **14** side is recessed in the longitudinal direction x. Each diameter of the first balloon **11** and the second balloon **12** becomes smaller as compared with the case where one balloon is used in place thereof, which contributes to an increase in curvature of the surface and a decrease in corners S**1** and S**2** generated at both ends of the case **15** when the first balloon **11** and the second balloon **12** are inflated, so that a space in the case **15** can be used efficiently.
(16) The first balloon **11** and the second balloon **12** are formed of an elastic member, and the material thereof is not limited to particular materials, but from the viewpoint of biocompatibility, for example, silicone rubber or the like is preferable. The first balloon **11** and the second balloon **12** are formed to be able to store therein water or the like to be supplied to organisms such as mice and are provided to be able to discharge the water or the like therefrom.
(17) The second balloon **12** has a larger contraction force than that of the first balloon **11**. The first balloon **11** according to the present embodiment is formed to have a full length longer than that of the second balloon **12**, that is, the first balloon **11** and the second balloon **12** are attached to the shaft **10** so that the first balloon **11** is longer than the second balloon **12** along the shaft **10** in the longitudinal direction x.
(18) Note that in the present embodiment, the second balloon **12** is formed to be shorter than the first balloon **11** in the longitudinal direction x so that the second balloon **12** has a larger contraction force than that of the first balloon **11**, but is not limited thereto. In the case where the contraction force of the second balloon **12** is set to be larger than that of the first balloon **11**, the first balloon **11** and the second balloon **12** may be made different from each other in thickness and inner diameter.
(19) The first balloon **11** has a first hole **11***a* formed at one end, and a second hole **11***b* formed at the other end. The shaft **10** extends through the first hole **11***a* and the second hole **11***b*. The first hole **11***a* and the second hole **11***b* are provided with shaft members **11***c* and **11***d*, respectively, the shaft members **11***c* and **11***d* each having a cylindrical shape protruding inwardly of the first balloon **11** from a corresponding end of the first balloon **11**. The first balloon **11** is formed to be in close contact with an outer peripheral surface of the shaft **10** extending therethrough using the shaft members **11***c* and **11***d*.
(20) Like the first balloon **11**, the second balloon **12** also has a first hole **12***a* formed at one end, and a second hole **12***b* formed at the other end. The shaft **10** extends through the first hole **12***a* and the second hole **12***b*. The first hole **12***a* and the second hole **12***b* are provided with shaft members **12***c* and **12***d*, respectively, the shaft members **12***c* and **12***d* each having a cylindrical shape protruding inwardly of the second balloon **12** from a corresponding end of the second balloon **12**. The second balloon **12** is formed to be in close contact with the outer peripheral surface of the shaft **10** extending therethrough using the shaft members **12***c* and **12***d*.
(21) As illustrated in FIG. **1**, the first balloon **11** and the second balloon **12** are attached to be adjacent to each other in the longitudinal direction x of the shaft **10**. In the present embodiment, the first balloon **11** is disposed on an injection port **10***a* side of the shaft **10**, and the second balloon **12** is disposed on a discharge port **10***b* side of the shaft **10**. Thus, the shaft member **11***d* of the first balloon **11** and the shaft member **12***c* of the second balloon **12** contact each other, and the first balloon **11** and the second balloon **12** are provided to expand and inflate in a direction away from the shaft **10** from between the shaft member **11***d* of the first balloon **11** and the shaft member **12***c* of the second balloon **12**. Thus, as the first balloon **11** and the second balloon **12** inflate, a region in which the first balloon **11** and the second balloon **12** are in close contact with each other increases around the shaft member **11***d* of the first balloon **11** and the shaft member **12***c* of the second balloon **12**. Note that FIG. **1** illustrates a state in which the first balloon **11** and the second balloon **12** are inflated, but in the state of contracting, each of the first balloon **11** and the second balloon **12** is in substantially close contact with the shaft **10**.

(22) The shaft **10** is a tubular member having a cylindrical shape. The shaft **10** extends from one end to the other end in the longitudinal direction x of the case **15**, and as illustrated in FIG. **2**, a flow path **10***e* in which the water flows is formed along the shaft center. One end of the shaft **10** has the discharge port **10***b* for discharging the water or the like stored in the first balloon **11** and the second balloon **12**. On the other hand, the other end of the shaft **10** has the injection port **10***a* for injecting the water or the like into the interior of the first balloon **11** and the interior of the second balloon **12**. The flow path **10***e* is formed to extend from the injection port **10***a* to the discharge port **10***b*.
(23) The injection port **10***a* according to the present embodiment is adapted to be connected with a one-way valve **40** for preventing the water or the like in the shaft **10** from flowing out from the shaft **10**, but a one-way valve may be incorporated into the injection port **10***a* itself. The injection port **10***a* is provided to be connected with an unillustrated pipe or the like via the one-way valve.
(24) The shaft **10** has a first communication hole **10***c* formed to directly connect an interior space of the shaft **10** formed into a cylindrical shape and the flow path **10***e* in an interior space of the first balloon **11**, thereby permitting fluid communication therebetween. In addition, the shaft **10** has a second communication hole **10***d* formed to directly connect an interior space of the shaft **10** formed into a cylindrical shape and an interior space of the second balloon **12**, thereby permitting fluid communication therebetween.
(25) The shaft **10** includes a connection portion **10***f* that connects an end of the first balloon **11** and an end of the second balloon **12**, the ends being adjacent to each other, or connects the shaft member **11***d* (see FIG. **1**) of the first balloon **11** and the shaft member **12***c* (see FIG. **1**) of the second balloon **12**, and the flow path **10***e* is also formed in the connection portion **10***f*. The connection portion **10***f* is formed into a cylindrical shape, and an outer peripheral surface of the cylindrical shape portion is formed to have an outer diameter enabling the shaft member **11***d* of the first balloon **11** to be in close contact with the shaft member **12***c* of the second balloon **12**. Note that the shaft **10** according to the present embodiment is formed into a cylindrical shape having a constant outer diameter from the first hole **11***a* of the first balloon **11** to the second hole **12***b* of the second balloon **12**, but is not limited thereto, and a portion between the shaft member **11***c* (see FIG. **1**) and the shaft member **11***d* (see FIG. **1**) of the first balloon **11** may have different outer diameters and different outer peripheral shapes. The fluid supply apparatus **100** has the connection portion **10***f*, thereby permitting fluid communication between the first balloon **11** and the second balloon **12** as well as fluid communication between the discharge port **10***b* and the injection port **10***a*.
(26) As illustrated in FIG. **1**, the case **15** is formed into a cylindrical shape formed of a transparent resin material. Note that the case **15** may be formed into not only a cylindrical shape but also any tube shape such as a tube shape having a polygonal cross-section as long as it has a tubular shape. The shaft **10**, the first balloon **11**, and the second balloon **12** are disposed in the case **15**. The first cap **13** is attached to the case **15** so that one opening of the case **15** is closed. Also, the second cap **14** is attached to the case **15** so that the other opening of the case **15** is closed. Accordingly, both ends of the shaft **10** protrude outward from both ends in the longitudinal direction x of the case **15**.
(27) Hereinafter, there will be described a procedure for storing water or the like in the fluid supply apparatus **100** and a procedure of discharging the water or the like from the fluid supply apparatus **100**. In order to store water or the like in the fluid supply apparatus **100**, an unillustrated liquid source is connected to the injection port **10***a* of the shaft **10** to inject the water or the like from the injection port **10***a* into the shaft **10**. In the present embodiment, the contraction force of the first balloon **11** is smaller than the contraction force of the second balloon **12**, and therefore, the water or the like injected into the shaft **10** is injected into the first balloon **11** through the first communication hole **10***c*. At this time, the first balloon **11** inflates according to an injection amount of water or the like.
(28) When the first balloon **11** is inflated until the outer peripheral surface thereof contacts the inner peripheral surface of the case **15**, the water or the like being injected into the shaft **10** is

injected into the second balloon **12**, whereby the second balloon **12** inflates according to the injection amount of water or the like.
(29) When the second balloon **12** is inflated until the outer peripheral surface thereof contacts the inner peripheral surface of the case **15**, the injection of the water into the fluid supply apparatus **100** is completed.
(30) As illustrated in FIG. **3**A, the fluid supply apparatus **100** according to the present embodiment is provided with a first pressure sensor **33** for detecting that the first balloon **11** has contacted the case **15**, a second pressure sensor **34** for detecting that the second balloon **12** has contacted the case **15**, and a third pressure sensor **35** for detecting an internal pressure in the flow path **10***e*, the third pressure sensor **35** being incorporated into the flow path **10***e* of the shaft **10**. A user of the fluid supply apparatus **100** can determine whether the outer peripheral surfaces of the first balloon **11** and the second balloon **12** have contacted the inner peripheral surface of the case **15**, on the basis of the detection results of the first pressure sensor **33** and the second pressure sensor **34**, and detect the pressure of the liquid stored in the fluid supply apparatus **100** on the basis of the detection result of the third pressure sensor **35**.
(31) When the water or the like is supplied to organisms such as mice, a valve of the unillustrated pipe or the like connected to the shaft **10** is opened. Then, the contraction forces of the first balloon **11** and the second balloon **12** cause the water or the like inside the first balloon **11** and the second balloon **12** to be discharged from the discharge port **10***b* into the pipe, whereby the water or the like can be automatically supplied to organisms such as mice.
(32) When the water or the like inside the first balloon **11** and the second balloon **12** is discharged from the discharge port **10***b*, the internal pressure of the first balloon **11** changes as shown in FIG. **3**B. In this figure, the vertical axis represents the internal pressure (kPa) and the horizontal axis represents the elapsed time (hr). Note that, in FIG. **3**B, the description is made using the internal pressure of the first balloon **11**, but is not limited thereto, and may be made using the internal pressure of the second balloon **12**.
(33) In order to discharge the water from the discharge port **10***b* of the fluid supply apparatus **100**, the water or the like stored in the second balloon **12** having a larger contraction force than that of the first balloon **11** is first discharged from the discharge port **10***b*. When the discharge of the water or the like stored in the second balloon **12** is completed and is switched to the discharge of the water or the like stored in the first balloon **11**, or when about 36 hours have elapsed in FIG. **3**B, the internal pressure of the first balloon **11** rises and a first peak P**1** occurs. Thus, the user of the fluid supply apparatus **100** can detect the first peak P**1** using the third pressure sensor **35** without visually observing the first balloon **11** and the second balloon **12**, thereby determining that the balloon from which the water or the like is discharged has switched from the second balloon **12** to the first balloon **11**.
(34) When the discharge of the water or the like from the second balloon **12** is completed, the fluid supply apparatus **100** starts to discharge the water or the like stored in the first balloon **11**. At this time, the internal pressure of the first balloon **11** decreases with the discharge of the water or the like, but when the discharge of the water or the like stored in the first balloon **11** is completed, or when about 72 hours have elapsed in FIG. **3**B, the internal pressure of the first balloon **11** slightly rises and a second peak P**2** occurs. The user of the fluid supply apparatus **100** can detect the second peak P**2** using the third pressure sensor **35** without visually observing the first balloon **11** and the second balloon **12**, thereby determining that the water or the like stored in the first balloon **11** would soon be exhausted. Note that, in the present embodiment, storage amounts of the first balloon **11** and the second balloon **12** are determined using the third pressure sensor **35**, but a decrease in the storage amount of the first balloon **11** may be determined using the first pressure sensor **33** and a decrease in the storage of the second balloon **12** may be determined using the second pressure sensor **34**.
(35) The fluid supply apparatus **100** according to the present embodiment comprises the first

balloon **11** that is formed of an elastic body to be capable of storing water and the like, the second balloon **12** that is formed of an elastic body to be capable of storing water and the like, and has a larger contraction force than the contraction force of the first balloon **11**, the shaft **10** that supports both of the first balloon **11** and the second balloon **12**, and the case **15** that houses the first balloon **11** and the second balloon **12** and defines inflation ranges of the first balloon **11** and the second balloon **12**, wherein the shaft **10** has the flow path **10***e* formed to permit fluid communication between the interior of the first balloon **11** and the interior of the second balloon **12**. Thus, each diameter of the subdivided first balloon **11** and second balloon **12** becomes smaller as compared with the case where one balloon is used in place thereof, which contributes to an increase in curvature of the surface and a decrease in corners S**1** and S**2** generated at both ends of the case **15** when the first balloon **11** and the second balloon **12** are inflated, so that a space in the case **15** can be used efficiently. This makes it possible to eliminate the need for a power supply source and store fluid efficiently.
(36) The fluid supply apparatus **100** according to the present embodiment in which the second balloon **12** has a larger contraction force than that of the first balloon **11** can control order in which the first balloon **11** and the second balloon **12** inflate and contract, so that the second balloon **12** contracts earlier than the first balloon **11** and the first balloon inflates earlier than the second balloon **12**.
(37) Furthermore, contracting a plurality of balloons in order enables the fluid supply apparatus **100** according to the present embodiment to monitor a gradual decrease in the water or the like stored in the fluid supply apparatus **100**.
(38) Also, the fluid supply apparatus **100** according to the present embodiment comprises the first balloon **11** having a small contraction force, which enables the balloons to inflate smoothly even when the fluid is injected at a low pressure. This enables the balloons to inflate more easily than the case where only one balloon is provided and to inflate to fill in the corners S**1** and S**2** without excessively increasing the pressure of the fluid, thereby reducing the loads to the balloons.
Second Embodiment
(39) FIG. **4** illustrates a perspective view of a fluid supply apparatus according to a second embodiment of the present invention, FIG. **5** illustrates a cross section of the fluid supply apparatus according to the second embodiment, FIG. **6** illustrates a schematic perspective view through the fluid supply apparatus in which balloons are inflated, and FIG. **7** illustrates a schematic sectional view of the fluid supply apparatus in which balloons are inflated. Note that, in the second embodiment, parts different from the first embodiment are described, and the configurations in the drawings, which are substantially the same as the first embodiment, are denoted with the same reference signs.
(40) As illustrated in FIG. **4**, a fluid supply apparatus **200** according to the present embodiment comprises a support plate (support member) **110** having a plate shape, and a pair of cases (casings) **115** and **115** formed to cover both surfaces of the support plate **110**.
(41) As illustrated in FIG. **5**, the cases **115** and **115** each have a shape inflated in the direction away from the support plate **110**, and a cavity is formed between the support plate **110** and each of the cases **115** and **115**. In the cases **115** and **115**, a first balloon **111** is attached to one surface **110***a* of the support plate **110**, and a second balloon **112** is attached to the other surface **110***b* of the support plate **110**, the second balloon **112** having a larger contraction force than that of the first balloon **111**.
(42) The support plate **110** has a flow path **110***e* formed to extend in an in-plane direction with respect to the surfaces **110***a* and **110***b*, and the flow path **110***e* is in fluid communication with the discharge port **10***b* and the injection port **10***a*. The flow path **110***e* branches near the center of the support plate **110** to form a branch path **110***f* extending in an out-of-plane direction with respect to the surfaces **110***a* and **110***b*, and the branch path **110***f* is in fluid communication with the interior of the first balloon **111** and the interior of the second balloon **112** in the surfaces **110***a* and **110***b*.
(43) The first balloon **111** and the second balloon **112** are formed into a flat plate shape so as to be

in close contact with the surfaces **110***a* and **110***b* of the support plate **110**, respectively, in the state in which the first balloon **111** and the second balloon **112** are not inflated. The first balloon **111** and the second balloon **112** are adapted to inflate in the directions away from the surfaces **110***a* and **110***b* of the support plate **110**, respectively, as illustrated in FIG. **6**, when the fluid is injected into the first balloon **111** and the second balloon **112** through the flow path **110***e* and the branch path **110***f*. As illustrated in FIG. **7**, the cases **115** and **115** are provided to define the inflation ranges of the first balloon **111** and the second balloon **112**, and when being inflated, the first balloon **111** and the second balloon **112** come into close contact with their respective inner surfaces of the cases **115** and **115** so that the cases **115** and **115** limit the inflation of the first balloon **111** and the second balloon **112**.

(44) Hereinafter, there will be described the inflation of the first balloon **111** and the second balloon **112**. When the first balloon **111** and the second balloon **112** inflate actually, the first balloon **111** first starts to inflate, and the first balloon **111** contacts the inner surface of the case **115**.

(45) When the first balloon **111** contacts the inner surface of the case **115**, the second balloon **112** next starts to inflate, and the second balloon **112** contacts the inner surface of the case **115** and stops inflating.

Third Embodiment

(46) FIG. **8** illustrates a front view of a fluid supply apparatus according to a third embodiment of the present invention, and FIG. **9** illustrates a sectional view of a balloon unit. Note that, in the third embodiment, parts different from the second embodiment are described, and the configurations in the drawings, which are substantially the same as the first embodiment, are denoted with the same reference signs.

(47) As illustrated in FIG. **8**, a fluid supply apparatus **300** according to the present embodiment comprises a first balloon unit **50***a* in which a first balloon **111** (see FIG. **9**) is housed, a second balloon unit **50***b* in which a second balloon is housed, the second balloon having a larger contraction force than that of the first balloon **111**, and a third balloon unit **50***c* in which a third balloon is housed, the third balloon having a larger contraction force than that of the second balloon. The balloons in the first balloon unit **50***a*, the second balloon unit **50***b* and the third balloon unit **50***c* are connected in parallel through a pipe **316**. Note that the first balloon unit **50***a*, the second balloon unit **50***b*, and the third balloon unit **50***c* have substantially the same configuration and the balloons differ from one another only in the contraction force, and therefore, the description will be made below using the first balloon unit **50***a*.

(48) The present embodiment is different from the second embodiment in that each of the first balloon unit **50***a*, the second balloon unit **50***b*, and the third balloon unit **50***c* houses only one balloon **111** as illustrated in FIG. **9**.

(49) When the fluid is injected into the pipe **316** and each of the balloons **111** in the first balloon unit **50***a*, the second balloon unit **50***b*, and the third balloon unit **50***c* inflates, the balloon **111** in the first balloon unit **50***a* first inflates until the balloon **111** in the first balloon unit **50***a* contacts the inner surface of the case **115**. When the inflation of the balloon **111** in the first balloon unit **50***a* is completed, the balloon **111** in the second balloon unit **50***b* next inflates until the balloon **111** in the second balloon unit **50***b* contacts the inner surface of the case **115**. When the inflation of the balloon **111** in the second balloon unit **50***b* is completed, the balloon **111** in the third balloon unit **50***c* next inflates until the balloon **111** in the third balloon unit **50***c* contacts the inner surface of the case **115**, and the injection of the fluid into the fluid supply apparatus **300** is completed.

(50) In order to discharge the fluid from the liquid supply apparatus **300**, the balloon **111** in the third balloon unit **50***c* first contracts to discharge the fluid into the pipe **316**. When the contraction of the balloon **111** in the third balloon unit **50***c* is completed, the balloon **111** in the second balloon unit **50***b* next contracts to discharge the fluid into the pipe **316**. When the contraction of the balloon **111** in the second balloon unit **50***b* is completed, the balloon **111** in the first balloon unit **50***a* next contracts to discharge the fluid into the pipe **316**.

(51) Hereinbefore, the present invention has been described using the embodiments, but the present invention is not limited thereto. For example, in the above-described embodiments, the cases **15** and **115** are formed of a transparent resin material, but as long as the cases **15** and **115** can have rigidity enough to withstand the pressure applied by the inflated balloons, a transparent material may be used for only a part of the cases **15** and **115** or the entire cases **15** and **115** may be formed of a non-transparent material, for example.

REFERENCE SIGNS LIST

(52) P**1** . . . First peak P**2** . . . Second peak S**1** . . . Corner **10** . . . Shaft (support member) **10***a* . . . Injection port **10***b* . . . Discharge port **10***c* . . . First communication hole **10***d* . . . Second communication hole **10***e* . . . Flow path **10***f* . . . Connection portion **11** . . . First balloon **11***a* . . . First hole **11***b* . . . Second hole **11***c* . . . Shaft member **11***d* . . . Shaft member **12** . . . Second balloon **12***a* . . . First hole **12***b* . . . Second hole **12***c* . . . Shaft member **12***d* . . . Shaft member **13** . . . First cap **14** . . . Second cap **15** . . . Case (casing) **33** . . . First pressure sensor **34** . . . Second pressure sensor **35** . . . Third pressure sensor **40** . . . One-way valve **50***a* . . . First balloon unit **50***b* . . . Second balloon unit **50***c* . . . Third balloon unit **100** . . . Fluid supply apparatus **110** . . . Support plate **110***a* . . . Surface **110***b* . . . Surface **110***e* . . . Flow path **110***f* . . . Branch path **111** . . . First balloon **112** . . . Second balloon **115** . . . Case (casing) **200** . . . Fluid supply apparatus **300** . . . Fluid supply apparatus **316** . . . Pipe

Claims

1. A fluid supply apparatus for supplying stored fluid to outside, comprising: a first balloon that is formed of an elastic body to be capable of storing the fluid; a second balloon that is formed of an elastic body to be capable of storing the fluid, and has a larger contraction force than a contraction force of the first balloon; a support member that supports both of the first balloon and the second balloon; and a casing that houses the first balloon and the second balloon, and defines inflation ranges of the first balloon and the second balloon, wherein the support member has a flow path formed to permit fluid communication between an interior of the first balloon and an interior of the second balloon, and wherein the support member is a tubular member and extends through the first balloon and the second balloon.
2. The fluid supply apparatus according to claim 1, wherein the support member includes a connection portion that connects an end of the first balloon and an end of the second balloon, the ends being adjacent to each other, and the flow path is formed in the connection portion.
3. The fluid supply apparatus according to claim 1, wherein the support member extends from one end to the other end of the casing.
4. The fluid supply apparatus according to claim 1, wherein the support member has a plate shape, and the first balloon is attached to one surface of the support member and the second balloon is attached to the other surface of the support member.
5. The fluid supply apparatus according to claim 1, wherein the casing is provided with a sensor for detecting that at least one of the first balloon and the second balloon has contacted the casing.
6. The fluid supply apparatus according to claim 1, wherein the first balloon and the second balloon are different from each other in thickness.
7. The fluid supply apparatus according to claim 1, wherein the first balloon and the second balloon are different from each other in inner diameter.
8. The fluid supply apparatus according to claim 1, wherein the first balloon and the second balloon are different from each other in length.
9. A fluid supply apparatus for supplying stored fluid to outside, comprising: a first unit that comprises a first balloon that is formed of an elastic body to be capable of storing the fluid, a first support member that supports the first balloon, and a first casing that houses the first balloon and defines an inflation range of the first balloon; and a second unit that comprises a second balloon

that is formed of an elastic body to be capable of storing the fluid and has a larger contraction force than a contraction force of the first balloon, a second support member that supports the second balloon, and a second casing that houses the second balloon and defines an inflation range of the second balloon, wherein the first support member and the second support member have a flow path formed to permit fluid communication between an interior of the first balloon and an interior of the second balloon.

10. The fluid supply apparatus according to claim 9, wherein the first casing is provided with a sensor for detecting that the first balloon has contacted the first casing.

11. The fluid supply apparatus according to claim 9, wherein the second casing is provided with a sensor for detecting that the second balloon has contacted the second casing.

12. The fluid supply apparatus according to claim 9, wherein the first balloon and the second balloon are different from each other in thickness.

13. The fluid supply apparatus according to claim 9, wherein the first balloon and the second balloon are different from each other in inner diameter.

14. The fluid supply apparatus according to claim 9, wherein the first balloon and the second balloon are different from each other in length.